

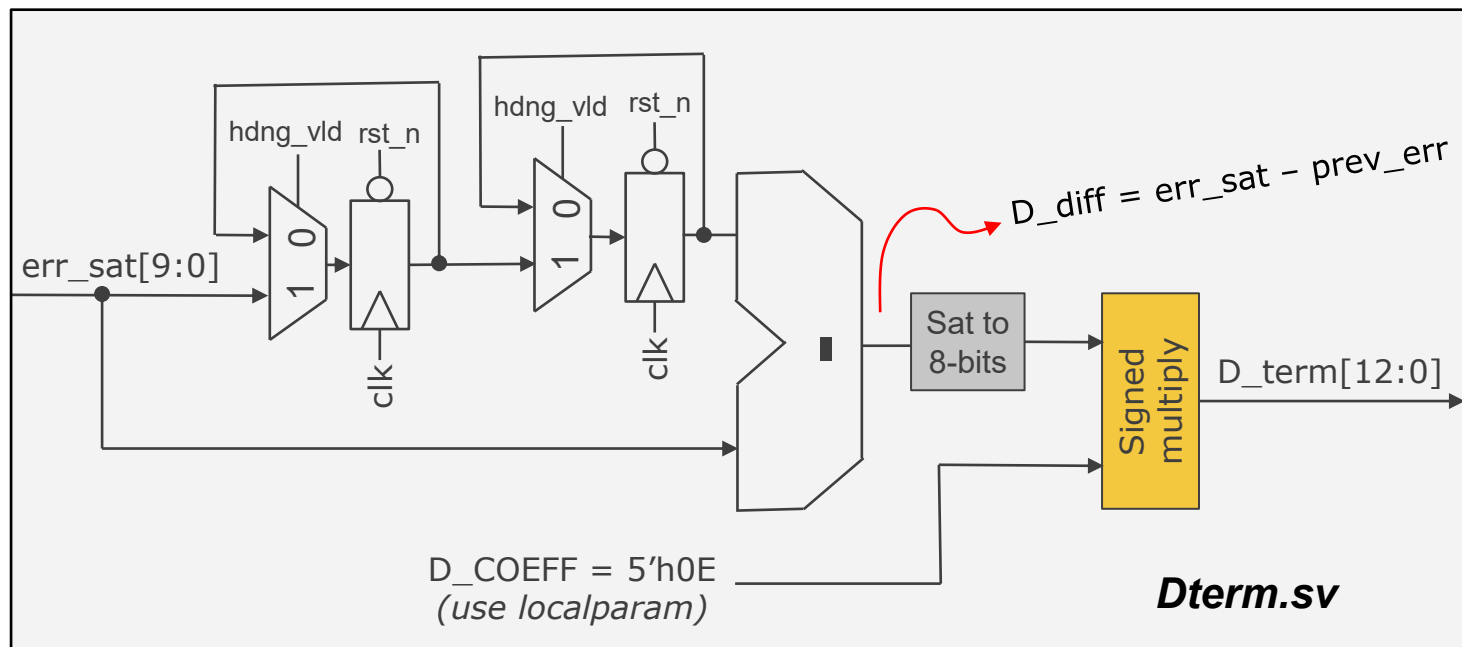


Exercise 10: D Term & PID Assembly

- The **derivative** of a function can be approximated by:

$$\frac{dF(t)}{dt} = \frac{f(t) - f(t - \Delta t)}{\Delta t}$$

- We only need something *proportional* to the derivative
→ divide by Δt is unnecessary

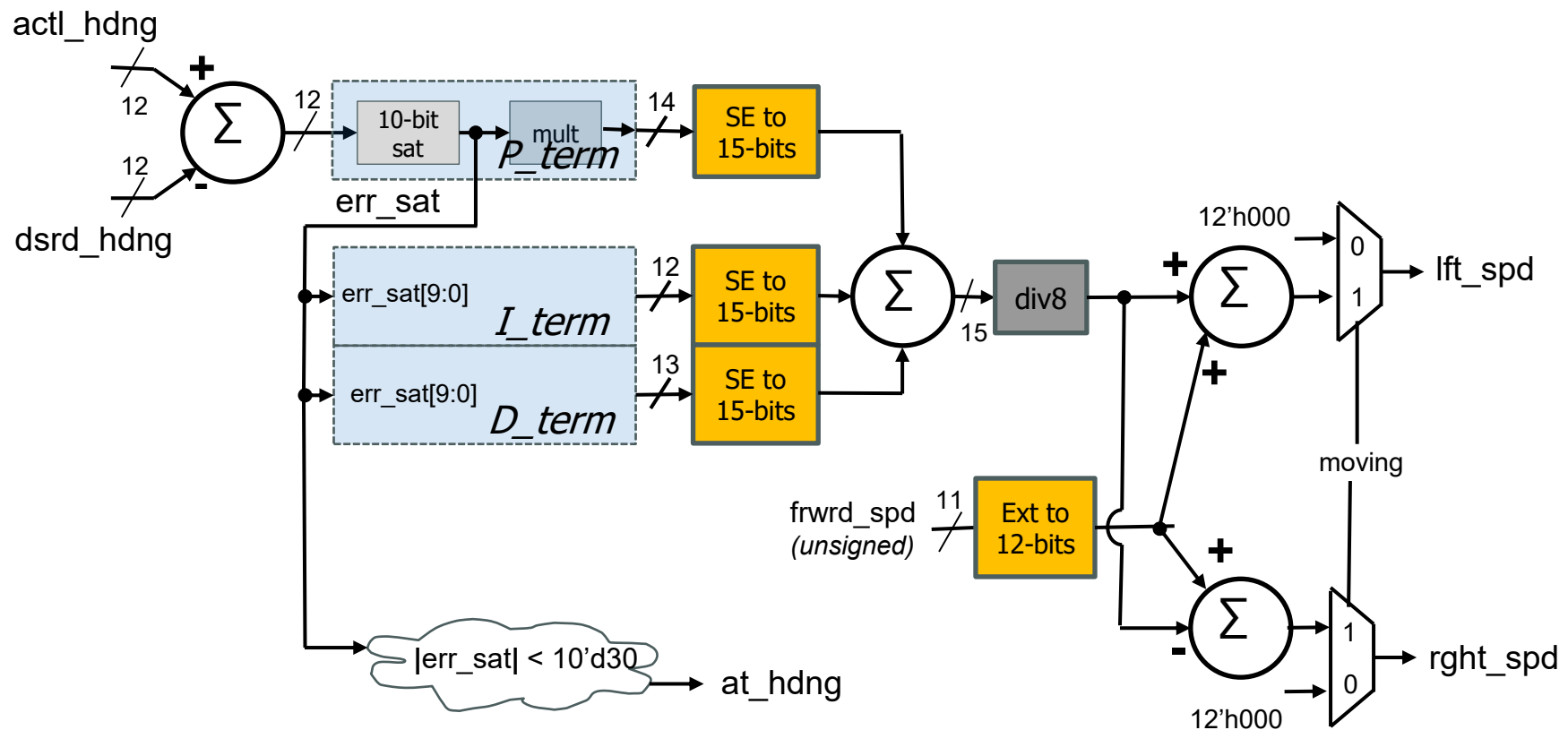


- Implement & submit: **Dterm.sv**, **Dterm_tb.sv**, proof testbench ran
- Testbench should be self-checking



Exercise 10: D Term & PID Assembly

- Now that you have P, I & D, assemble the PID unit



- NOTE:** The blocks ***P_term***, ***I_term***, & ***D_term*** are shown outlined in dotted lines because you may not want to instantiate them, but rather copy their implementation in flat into **PID.sv**.



Exercise 10: D Term & PID Assembly

- PID.sv interface

Signal	Dir	Description
clk, rst_n	in	50MHz clock and asynch active low reset
moving	in	High if turning or moving forward (from navigate block)
dsrd_hdng[11:0]	In	Signed desired heading (from cmd_proc)
actl_hdng[11:0]	in	Signed actual heading (from inert_intf)
hdng_vld	In	High for 1 clock every new valid gyro reading
frwrd_spd[10:0]	In	Unsigned forward speed (from navigate block)
at_hdng	out	Asserted if error is small
lft_spd[11:0]	out	Signed left motor speed (goes to MtrDrv)
rght_spd[11:0]	out	Signed right motor speed

- Create **PID.sv**
- A basic self-checking testbench is provided (**PID_tb.sv**)
- Submit **PID.sv** and an image of the transcript window showing it passes the provided testbench